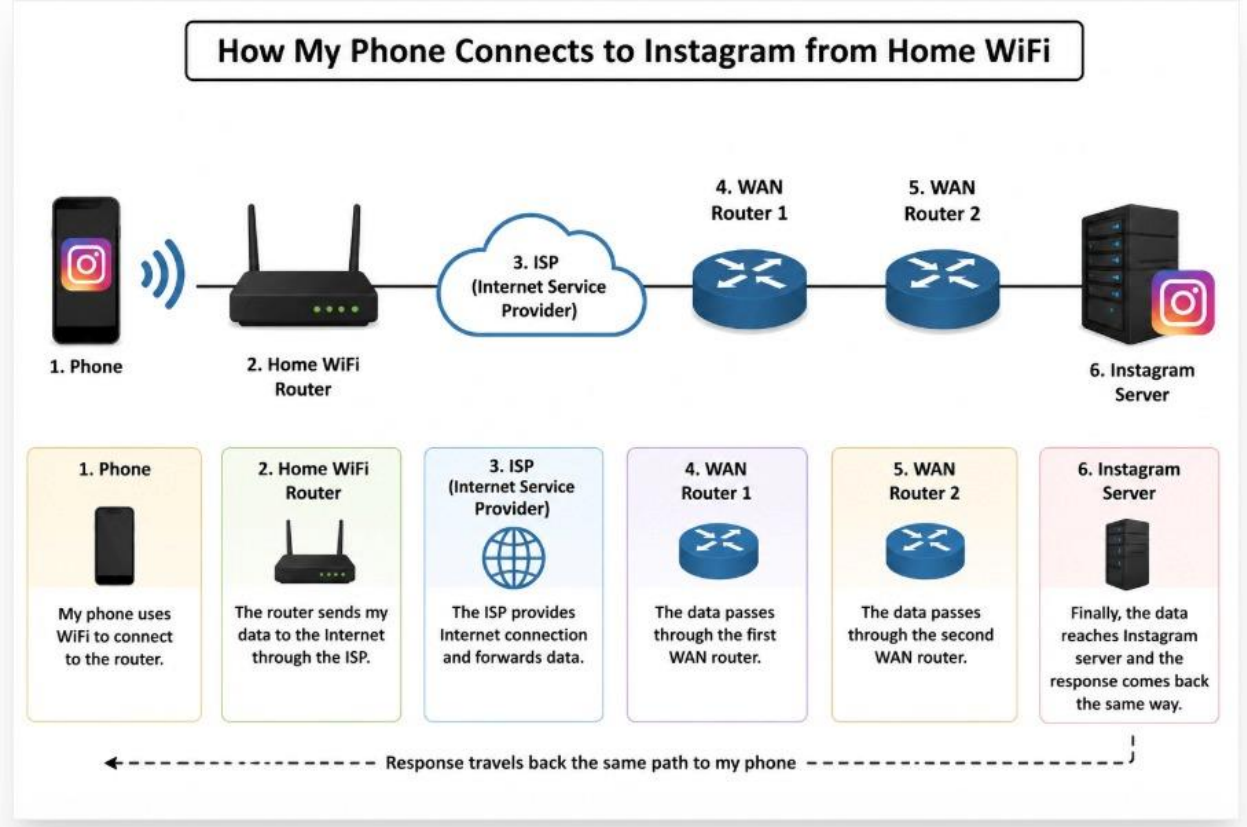
1.Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.



2.Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

WAN Technology	Speed	Cost	Real-Life Example
MPLS (Multiprotocol Label Switching)	High (10 Mbps to 10 Gbps)	High	Used by large companies such as a Zomato office to securely connect multiple branches.
Leased Line	Very High and Dedicated (100 Mbps to several Gbps)	Very High	Used by organizations such as IRCTC ticketing systems, banks, and data centers for reliable connectivity.
Broadband Internet	Medium to High (50 Mbps to 1 Gbps)	Low	Commonly used in homes for Wi-Fi, streaming, online classes, and social media.

3.Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.Hint: Focus on how the VPN creates a private tunnel over the public WAN.

VPN and WAN Explanation

A VPN (Virtual Private Network) allows a user to connect securely to the internet even when using a public Wi-Fi network, such as the Wi-Fi in a coffee shop. The coffee shop Wi-Fi is connected to the internet through a Wide Area Network (WAN).

When you turn on a VPN, it creates an encrypted private tunnel between your phone or laptop and a VPN server on the internet. All the data you send—such as opening Spotify or YouTube—travels through this secure tunnel before reaching the app's servers. Because the data is encrypted, other people on the same public Wi-Fi network cannot easily read your passwords, messages, or browsing activity.

In simple terms, a VPN uses the public WAN (the internet) but protects your information by creating a private encrypted connection across that public network. This makes it much safer to access apps and websites while using public Wi-Fi in places like coffee shops, airports, or hotels.

4.Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

For Flipkart, the best WAN technology would be MPLS (Multiprotocol Label Switching).

Recommended WAN Technology: MPLS

Reason 1: Secure communication MPLS provides a private and secure network connection between warehouses in Ahmedabad, Mumbai, and Delhi. This helps protect inventory data, order information, and internal company communications.

Reason 2: Reliable and fast performance MPLS offers high-speed, low-latency, and reliable connectivity, which is important for real-time warehouse management, order processing, and coordination between multiple locations.